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Pulse-driven MEMS gas sensor combined with machine learning for selective gas identification

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Abstract

The sensing and identification of trace gases are essential for ensuring chemical safety and protecting human health. This study introduces a low-power electronic nose system that utilizes a single sensor driven by repeated pulsed power inputs, offering a viable alternative to conventional sensor array-based methods. The sensor's compact design and suspended architecture facilitate a rapid thermal response, effectively decoupling the influences of temperature, physisorption, and charge exchange on the conductivity of the sensing material. This mechanism generates distinct gas sensing responses, characterized by alternating dual responses within a single time period. The unique dynamics of the dual signals, which vary with gas type and concentration, enable precise identification of multiple gas species using machine learning (ML) algorithms. Microfabricated through wafer-level batch processing, our innovative electronic nose system holds significant potential for battery-powered mobile devices and IoT-based monitoring applications.

Introduction

The increasing demand for cost-effective and energy-efficient gas sensors is driven by the rapid growth of wireless applications across various monitoring platforms, including stationary systems, and wearable devices^{1,2}. Recently, these sensors have expanded applications in diverse fields, such as indoor air quality monitoring, food quality assessment, industrial safety, explosive detection, and early disease diagnosis through breath analysis³. Conventionally, these tasks relied on expensive, bulky, and power-intensive devices, primarily using optical or electrochemical gas sensors⁴. Metal oxide semiconductor (MOS) gas sensors, however, offer compelling advantages, including simple fabrication, small size, reasonable operational lifespan, and excellent compatibility with electronic components^{2,5}. MOS gas sensors detect variations in electrical conductivity upon exposure to target analytes⁶, and they are well-suited for highly integrated sensor systems⁷. Despite these advantages, MOS sensors still face major challenges, including high power consumption and poor selectivity, which limit their practical

employment in portable electronics and wireless applications⁸.

High power consumption in MOS sensors arises from the activation energy required for gas adsorption and desorption on the MOS surface. Efforts to address this issue include using room-temperature sensing materials^{9,10}, incorporating miniaturized Joule heaters^{11,12}, and employing photoactivation techniques¹³. Although room-temperature sensing reduces power consumption, it suffers from poor recovery rates due to insufficient thermal energy for effective gas desorption^{9,14}. Joule-heating methods heat the sensing material to 100–400 °C, facilitating desorption and enhancing sensor performance¹⁵. The sensor response is influenced by two factors: conductivity changes due to temperature variations and surface charge exchange induced by target gases (i.e., chemisorption)¹⁶, leading to a coupled sensor output signal. To ensure stable monitoring with well-controlled temperature distribution at low power, T. Li and co-workers^{17–21} pioneered the microelectromechanical systems (MEMS) heaters, such as the suspended two-beam bridges and an impressive 3D microheater that enhance thermal efficiency by reducing heat loss through convection and providing thermal insulation from the substrate. In addition, pulsed heating, which periodically toggled the

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heater between power-on and low-power states, has been rigorously explored as a power-saving strategy, minimizing the active operation time and average power consumption^{22–24}.

Another major challenge for MOS sensors is gas cross-sensitivity, which impairs selectivity by causing responses to multiple gas types. Strategies to improve selectivity include adding porous overlayers as molecular sieves³, surface functionalization with noble metals or catalytic oxides^{25–27}, combining sensors into arrays^{28–30}, employing advanced data analysis techniques³¹, and using temperature modulation³². While molecular sieves can discriminate gases based on diffusion properties, they require frequent replacement and struggle to detect trace molecules due to strong adsorption. Electronic nose systems enhance selectivity by integrating multiple sensors and analyzing responses using ML algorithms³³. However, this approach paradoxically increases system complexity, power consumption, and cost³⁴. Thus, achieving effective gas discrimination with a single sensor remains a priority. Temperature modulation has been widely used to extract kinetic information owing to its ease of operation^{35,36}. In addition, it facilitates the extension of data obtained from isothermal resistance measurements of MOS sensors.

Alternative to operating under isothermal heating mode, pulsed heating has shown promise in gas discrimination by providing transient response data related to gas diffusion and gas–solid surface reactions^{37,38}. While pulse-driven MOS sensors have primarily focused on reducing power consumption, and enhancing sensitivity³⁸ and recovery³⁹, their potential for gas identification through multivariable feature analysis has not been explored extensively.

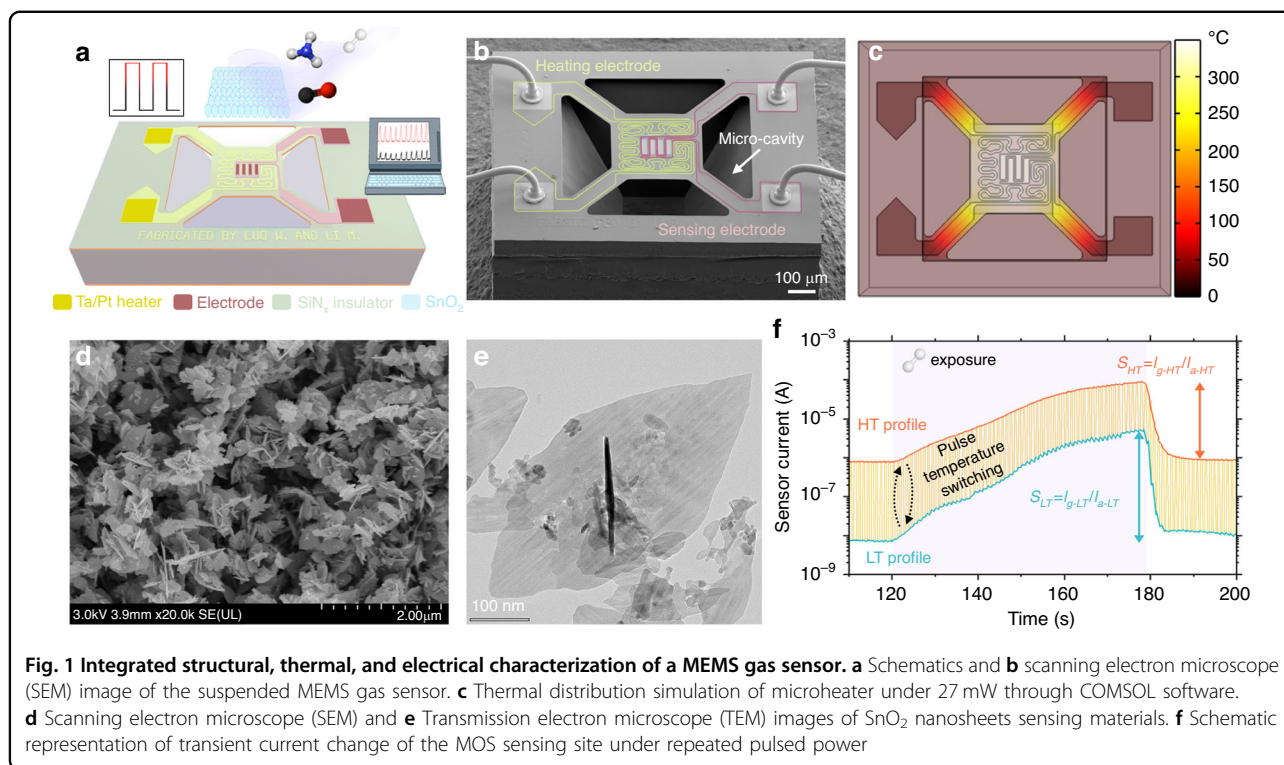
This study presents a dual-mode microheater-based MOS gas sensor that addresses both high power consumption and limited selectivity through pulsed heating mode (PHM). Fabricated using wafer-level processes, the sensor integrates SnO₂ nanosheets on a suspended MEMS microheater, enabling rapid heating within 20 ms due to its miniaturized, thermally insulated architecture. This design allows the sensor to generate distinct dual gas-sensing features within a single time domain, eliminating the need for complex multisensory setups and reducing power demands. Herein, the transient signals from a pulsed temperature sensor were employed. The extended datasets, both coupled with and decoupled from the thermally-induced response features, were evaluated via principal component analysis (PCA) and ML algorithms, including linear discriminant analysis (LDA), k-nearest neighbors (KNN), support vector machines (SVM), and random forests (RF). These models achieved 100% accuracy in identifying between three single-component gases (H₂, CO, and NH₃) based on transient response characteristics related to gas diffusion and

surface reaction kinetics. This single-sensor-based approach offers significant potential for advancing low-power, high-performance electronic nose systems and wearable IoT applications with gas discrimination capabilities.

Experimental

As depicted in Figs. 1a and S1a, the microheater platform was fabricated through three primary wafer-level microfabrication steps⁴⁰: forming a suspended supporting layer, integrating the microheater and sensing electrode layers, and depositing an insulating film. The microheater and detecting electrodes are positioned in parallel on the same metal layer, with the microheater located on the exterior and the sensing electrodes on the interior. This arrangement permits insulation between the two components without the need for an isolating silicon nitride layer. Meanwhile, this design simplifies the manufacturing process and reduces fabrication costs. Patterning was carried out using a conventional mask aligner (SUSS MA/BA6, Germany). Initially, a 1 µm-thick silicon nitride supporting layer was deposited on a 4-inch silicon wafer via low-pressure chemical vapor deposition (LPCVD). The deposition process employed SiH₂Cl₂ and NH₃ gases, with flow rates of 100/20 sccm. This process was carried out at 835 °C to minimize residual stress. Next, tantalum/platinum heating and sensing electrodes (15/200 nm in thickness) were selectively deposited onto the silicon nitride layer using photolithography and electron-beam evaporation. Platinum is picked as the electrode owing to its exceptional chemical and thermal stability. Following the lift-off of the photoresist layer, a 500 nm-thick insulating silicon nitride insulating layer was coated via plasma-enhanced chemical vapor deposition (PECVD) utilizing SiH₄, NH₃, and N₂ gases at 350 °C. Subsequently, an 8 µm-thick positive photoresist (P4620, AZ Electronic Materials, USA) was spin-coated onto the wafer and soft-baked. Openings for the sensing electrodes, bonding pads, and the etch window for the silicon concave pit were patterned through inductively coupled plasma (ICP) etching using Ar and SF₆ plasma at flow rates of 20/80 sccm. Notably, the exposed platinum surfaces functioned as a protective hard mask, protecting the underlying support layer during the etching process. The suspended structure was released by etching the silicon substrate with a 60 °C potassium hydroxide solution for 12.5 h, while a trench was created. The 4-inch wafer was subsequently diced into chips, which were then bonded to ceramic packages for further testing.

The temperature profile of the suspended microheater was simulated using COMSOL Multiphysics 6.0 (Fig. 1c). The heat transfer and electric current modules were coupled in the simulation, with convection and radiation effects neglected because of the microheater's relatively



low operating temperature and the low Rayleigh number associated with its small size. A 3 V electric potential was applied to the microheater under standard atmospheric conditions (1 atm, 293 K).

SnO₂ nanosheets were synthesized via a facile hydrothermal method⁴¹. A solution containing 0.6 mmol SnCl₂·2H₂O and 8 mmol urea in 20 ml deionized water was maintained at 180 °C for 8 h. The resulting products were centrifuged, repeatedly washed with deionized water and ethanol to eliminate any residual ions, followed by drying at 80 °C overnight. Finally, the nanosheets were annealed at 400 °C for 2 h in air. The sensing material was dispersed in a 2-propanol aqueous solution (5 mg/mL) by sonication for 15 min. In accordance with previous studies^{42,43}, 1 μL droplets of the suspension were deposited onto the microheater using a micropipette, while the platform was heated to 100 °C and retained at that temperature for 2 h to aid solvent evaporation and ensure uniform material coating. To ensure the reliability of the experimental results, the sensors were subjected to aging at 290 °C for 5 days prior to testing. The sensor's morphology was analyzed using SEM (Hitachi SU8010, Japan) and TEM (FEI Tecnai G2 F30, USA).

In the gas sensor testing, the concentration of target analyte was precisely controlled by mixing a standard gas (1000 ppm in N₂) with dry air using mass flow controllers (MFCs). Two MFCs were utilized: MFC 1 regulated the injection of the target gas, while MFC 2 provided compressed air for carrying and diluting the gas. The SnO₂

sensing material was heated via a suspended microheater, activated by applying a square-wave current. Simultaneously, the diluted target analyte was pumped into a custom-designed gas testing chamber, and the variation in SnO₂ conductivity was recorded over time using a source meter. The pulse voltage was set to 3 V, with pulse durations ranging from 0.5 to 1 s, and reset to 1 V within the same interval, resulting in duty cycles of 1/3 (PHM 1, $t_{HT} = 0.5$ s, $t_{LT} = 1$ s), 1/2 (PHM 2, $t_{HT} = 0.5$ s, $t_{LT} = 0.5$ s), and 2/3 (PHM 3, $t_{HT} = 1$ s, $t_{LT} = 0.5$ s).

PCA, an unsupervised learning technique, was employed for dimensionality reduction. While supervised learning algorithms such as LDA, KNN, SVM, and RF were applied for gas classification. PCA and LDA are both dimensionality reduction techniques. PCA identifies the axis that maximizes the variance across different gas type classes. Through the application of an orthogonal transformation, PCA transforms a set of correlated variables into linearly uncorrelated variables, known as principal components⁴⁴. In contrast, LDA is a prevalently utilized supervised classification method that creates discriminant functions by linearly combining labeled data. This approach aims to maximize the variance between categories while minimizing the variance within categories⁴⁵. The KNN algorithm classifies data points by a majority vote of the k-nearest neighbors, based on the shortest Euclidean distance to the unknown point⁴⁶. SVM operates as a linear classifier in a high-dimensional feature space, generated by mapping the original n-dimensional input

vector into a K -dimensional space (where $K > n$) using a non-linear function⁴⁷. To ensure a robust comparison of algorithmic performance, the dataset was split into training and testing subsets. To address potential overfitting and data bias, k -fold cross-validation was implemented during model training. The input matrix (Table S1) comprised 12 features, including gas response and response time (t_{res}) under dual mode across three distinct PHMs. All training, testing, and validation procedures for the ML models were conducted in Python, with PyCharm serving as the primary development environment.

Results and discussion

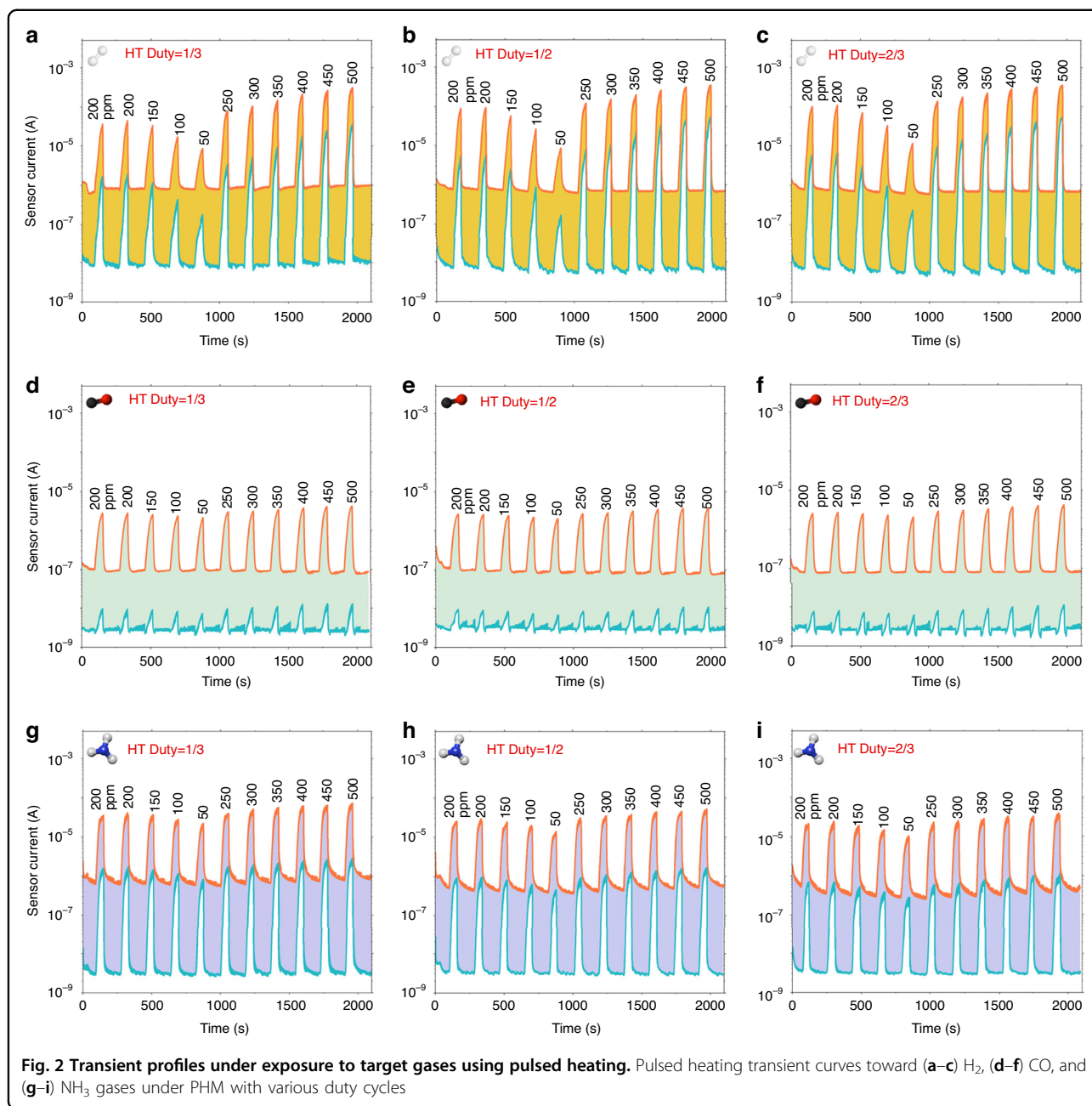
The temperature coefficient of resistance (TCR) procedure was employed to evaluate the operational temperature of the microdevice at a specific voltage. The heater integrated into the sensor shows a linear TCR. Electrical resistance was monitored using a source meter while heating the device in a temperature-controlled furnace. As revealed in Fig. S1b, the experimentally determined TCR for the microheater is $\sim 2.6 \times 10^{-3} \text{ }^{\circ}\text{C}^{-1}$. To examine the relationship between power consumption and work temperature, the sensor's electrical properties were analyzed in combined with the TCR (Fig. S1c). This method has been identified as an effective and convenient approach for assessing the temperature of a metal resistor⁴⁸. A prior thermo-mechanical study⁴⁹ demonstrated that the microheater can reach a maximum operational temperature of $650 \text{ }^{\circ}\text{C}$. However, when SnO_2 microdroplets are incorporated into the sensor, the maximum temperature decreases to $590 \text{ }^{\circ}\text{C}$ due to the increased thermal stress, which can lead to membrane fracture. In this study, optimizing the stress within each layer during the fabrication process results in minimal thermal mismatch stress between the multi-layers. Thus, the silicon nitride plate remains flat and crack-free following a pulsed-temperature heating experiment with a HT of $\sim 350 \text{ }^{\circ}\text{C}$, indicating its mechanical stability.

In this work, the suspended microheater (Fig. 1b) was fabricated using wafer-level batch processes, as described in the previous section. A $\sim 300 \text{ }\mu\text{m}$ silicon trench is observed beneath the suspended membrane following anisotropic wet etching. SEM and TEM images of SnO_2 nanosheets synthesized via a hydrothermal method are shown in Fig. 1d, e. These nanosheets exhibit a cross-linked, flake-like network with semi-open spaces, characterized by an average diameter of $\sim 127 \text{ nm}$ and an average thickness of $\sim 7 \text{ nm}$. This structure maximizes the exposed surface area, enhancing gas adsorption and improving sensitivity to low concentrations of target gases. Although MOS sensing materials were manually deposited onto the suspended micro-scale sensors (following the standard procedures used in commercial sensor manufacturing), these sensors exhibit moderate

uniformity and reproducibility in sensing performance, with a response relative standard deviation ($\text{RSD} < 10\%$), as previously reported in our earlier work²⁷. As gas sensor dimensions continue to decrease, the traditional manual deposition of sensing materials at specific locations becomes increasingly challenging. One potential solution is the implementation of a “top-down” approach for batch sensor fabrication, where MOS materials are dispersed in a photoresist. Micropatterning may then be realized by applying a MOS-photoresist suspension using a spraying, followed by photolithography process. Finally, a calcination step is required to remove any photoresist post to development. These concepts will be further investigated in future endeavors.

The serpentine-configured heating electrode ensures uniform temperature distribution across the sensing site, with a temperature gradient of less than $0.15 \text{ }^{\circ}\text{C}/\mu\text{m}$. The time required for temperature changes in the heater generally decreases as body size reduces, thereby enhancing the performance of MOS gas sensors that incorporate miniaturized heaters⁵⁰. Additionally, reducing the size of the active area of a microheater significantly lowers power consumption⁴⁸. Nevertheless, most biosensors and chemical sensors necessitate a larger active heated area to realize improved sensitivity⁴⁸, creating a persistent trade-off for researchers: balancing performance enhancements with practical constraints, such as energy efficiency. The proposed sensor platform features a central functional area of $278 \times 278 \text{ }\mu\text{m}^2$, heating up and cooling down rapidly within a voltage application period of just tens of milliseconds, which is comparable to other silicon-based microheaters reported in the literature^{18,51}. Owing to its compact size and excellent thermal insulation, the microheater demonstrates outstanding power efficiency ($13 \text{ K}\cdot\text{mW}^{-1}$), as indicated by a temperature increase of $\sim 350 \text{ }^{\circ}\text{C}$ achieved with only 27 mW of heater power. The gas sensor presented in this study offers notable advantages in terms of power and compactness. A thorough investigation into the impact of dimensional variations on sensor performance, along with a careful optimization of these factors, is another intriguing study beyond the scope of this work and will be addressed in subsequent studies.

Inspired by the dynamics of gas-solid interaction induced by thermal activation energy, a pulsed heating voltage (V_H) is applied to the heater to regulate the temperature. Thus, power consumption can be further reduced, whereas the power usage of the microheater-based sensor accounts for $\sim 30\%$ of the total energy typically required for Bluetooth communication in wireless monitoring systems. In addition to the response-recovery curves observed at high temperature (HT) during the heater-on state, distinct and transient gas sensing behaviors were also identified under low temperature (LT) pulses, which were decoupled from the thermally



enhanced electrical conductivity of the MOS material. These behaviors were observed in a cycled power input (Fig. 1f), and are attributed to the significantly faster thermal response compared to gas adsorption and desorption processes on the MOS surface. Figure 2 presents the transient current-time signals from SnO₂ nanosheets during microheater operation in PHMs. Upon initiation of heating, a 70-fold increase in conductivity was observed, which can be attributed to the negative temperature coefficient of resistance of the MOS material. The rapid thermal responses enable swift transitions between HT and LT phases, ensuring stable sensor operation at low

duty cycles while simultaneously reducing power consumption. However, there are limitations in reducing the duty cycle further, as the amount of data collected within a brief period is constrained by the measurement equipment. Herein, the heating pulse (0.5 s) was sufficiently large to ensure a data acquisition time of 0.1 s, thereby allowing for at least five data points to be collected in the pulsed state at the smallest duty cycle.

Upon exposure of the n-type SnO₂ surface to reducing gases such as H₂, CO and NH₃, electrons trapped by adsorbed oxygen species are released through gas chemisorption, causing an increase in the conductivity of the

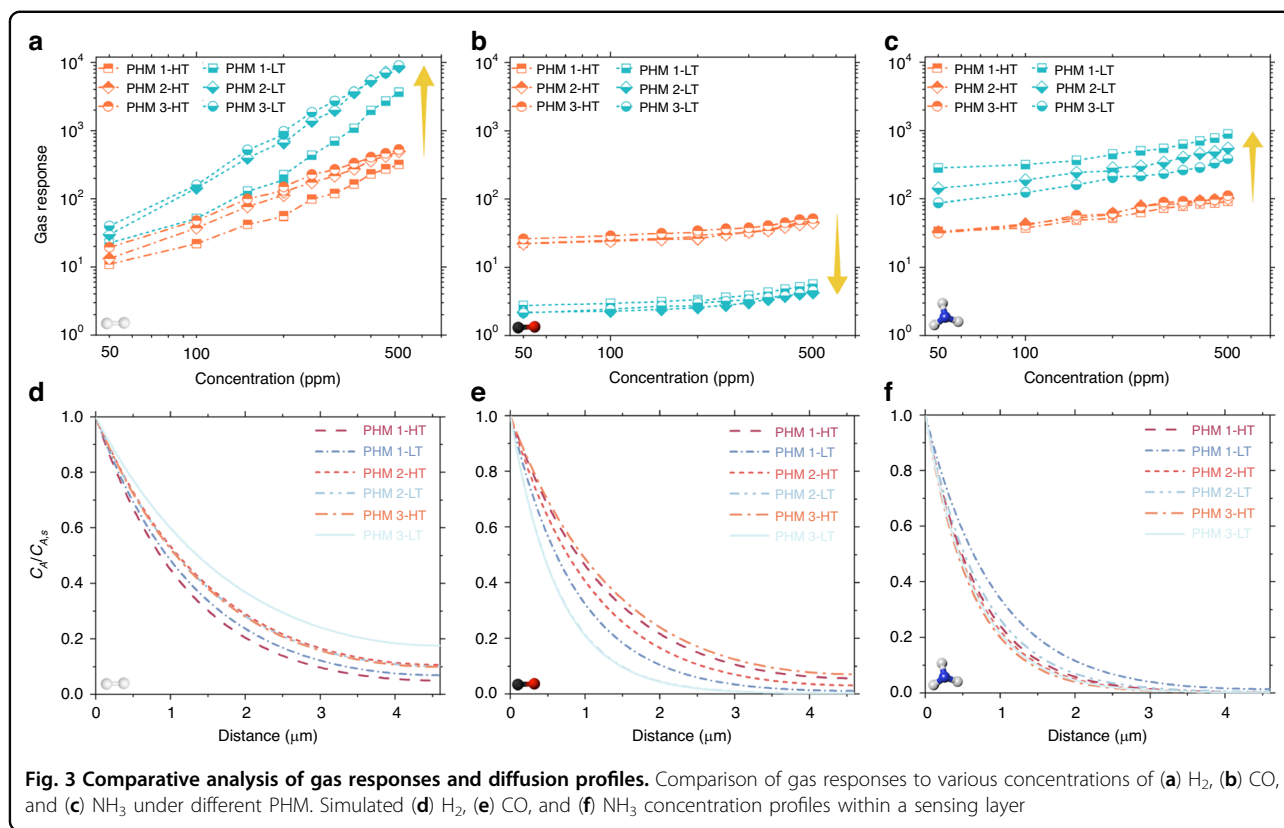
MOS⁵². The gas response for reducing gases is quantified as the ratio I_g/I_a , where I_g and I_a denote the current of gas sensor in the presence of the target analyte/air mixture, and in air, respectively. The repeated responses of the gas sensors with error bars (Fig. S2), revealed consistent reproducibility in detecting 250 ppm H_2 , CO , and NH_3 , respectively, with no significant variations during the testing period. This indicates the sensors' reliability in retaining stable operation over repeated usage. The long-term stability of SnO_2 nanosheet-based gas sensors has been extensively explored in the literature^{26,52,53}, as well as our previous work¹⁶, where the sensor responses demonstrated an RSD of less than 7% after a 150-day evaluation period.

Chemisorption requires a sufficient energy to activate the process, typically provided by heating the MOS materials. Consequently, changes in MOS conductivity upon gas exposure are influenced by both chemisorption and the thermally enhanced conductivity. However, under PHM, gas responses are observed not only during the heating phase but also during the cool-off phase. The repeated stabilization of conductivity signals in both states suggests that temperature fluctuations occur rapidly enough to mitigate the chem-desorption of oxygen species from the MOS surface. To gain insight into the gas response during the cool-off state, it is important to consider how pulse-induced temperature variations affect the adsorption, oxygen species, ion exchange, desorption of gas analytes, and the electrical properties of the sensing channel⁵¹. The charge interactions between gas analytes and MOS arise primarily from the chemisorption, a process that may also involve reactive oxygen species. In contrast, physisorption may not contribute to any charge exchange⁵⁴. As indicated in Fig. S3, prior to pulsed heating, the MOS surface is largely covered by a dense layer of pre-adsorbed air molecules (composed of a chemisorbed oxygen layer along with a thicker physisorbed layer⁵⁵), which prevents direct interaction between the gas molecules and the MOS surface, resulting in minimal electrical response at ambient temperature, particularly in the absence of prior thermal surface cleaning. The application of pulse heating facilitates the desorption of a thick air layer, whereas it is essential for enhancing the successive physisorption and chemisorption of target gases. When the MOS is subjected to HT under pulsed heating, this facilitates the chemisorption of both the target analytes and oxygen, as it provides the necessary activation energy ($E_{a-chem-adsorption}$)⁵⁶. During the chemisorption process, electrons are withdrawn from the reducing gas species, causing a change in MOS conductivity. Simultaneously, partial physical desorption of the analytes inevitably occurs, depicted by the activation energy ($E_{a-phys-desorption}$) and the corresponding Lennard-Jones potential curve (Fig. S3). During rapid pulsing cycles, abrupt temperature

fluctuations disrupt both adsorption and desorption processes³⁷. Consequently, PHM effectively mitigates the partial desorption of target gases. As the temperature of the MOS decreases intermittently during pulsed cycling, the target gas molecules diffuse and condense on the surface, as most of the pre-chemisorbed molecules are trapped within a potential well^{51,57,58}. This entrapment occurs because their chem-desorption requires a relatively high desorption activation energy ($E_{a-chem-desorption}$)⁵⁹. In other words, once chemisorption is activated, the rapid temperature variations in pulsed-temperature mode prevent the desorption of oxygen ions from the MOS surface⁵¹. This phenomenon is indirectly evidenced by the transient evolution of conductivity under a LT pulse. Meanwhile, physisorption may occur spontaneously, effectively stimulating the transition of analyte molecules and oxygen species into chemisorbed states while also providing additional sources for chemisorption in the subsequent HT pulse. Therefore, the intermittent cool-off phase eases the accumulation of chemisorption, which may be challenging when using isothermal measurements.

Temperature significantly influences both the surface charge exchange and the electrical properties of the MOS sensing channel. When gases chemisorb on the MOS, the resulting carrier variation may remain constant. However, at LT, the number of thermally activated carriers decreases, leading to reduced baseline conductivity (I_a) in the sensing channel. This reduction, in turn, amplifies the electrical response readout (I_g/I_a) at LT. Analyzing the effects of temperature on gas adsorption and desorption at the MOS surface and the conductivity of the sensing channel suggests that PHM provides a generalized framework for decoupling the various steps involved in gas molecule sensing. The synergetic effects coupled with persistent reactive oxygen species on the MOS surface facilitate the boosted measurement of conductivity changes at LT pulses, which are primarily due to chemisorption and are decoupled from thermally induced changes in conductivity. Both HT and LT reactions were observed within a single time frame, as illustrated by the orange (HT or coupled data) and cyan (LT or decoupled data) curves in Fig. 2. Given that the binding states of various adsorbed gas molecules differ across specific MOS surface facets and exhibit distinct thermodynamic properties, the pulsed temperature-modulated signals likely comprise thermodynamic and kinetic information specific to each gas molecule. Consequently, different analyte molecules may exhibit distinct activation energy constants for $E_{a-chem-adsorption}$, $E_{a-phys-desorption}$, and $E_{a-chem-desorption}$ on a given MOS sensor surface. These variations give rise to distinct temperature-dependent sensing behaviors, thereby resulting in a unique response pattern.

The oxidation of reducing species occurs through distinct reactions with reactive oxygen species, leading to



variations in gas adsorption/desorption patterns and electron transport, depending on the gas type⁵⁰, and influenced by the differences in electron affinity values between different gases (2.0 eV for H_2 ⁶⁰, 1.3 eV for CO ⁶¹, and 0.2 eV for NH_3 ⁶²). These variations lead to subtle differences in sensing kinetics. In addition, temperature changes can affect both the rate of adsorption and desorption, as well as the activation energy required for these processes. This effect arises from the interplay between surface reactivity and the diffusion of gas molecules^{51,63}. Following a pulse switch, a new equilibrium must be created, and the relaxation from a non-equilibrium to an equilibrium state is strongly dependent on the specific gas species present in the ambient environment^{64,65}. Hence, the gas response and its transient behavior under HT and LT pulses may vary based on the type of gas, concentration, and PHM. These distinctive traits indicate that analyzing pulse-driven gas responses could serve as an innovative approach to improving gas selectivity and discrimination.

Building upon this, the gas categorization capability of pulse-driven sensor operation was assessed. Figure 3a–c illustrates the gas responses to H_2 , CO , and NH_3 as a function of concentration, comparing the HT and LT phases. As expected, gas responses increase with higher concentrations, demonstrating a direct correlation between gas concentration and sensor response.

However, the responses were not sufficiently distinct to differentiate between the three gases. Although sensor current values were smaller in the LT phase compared to the HT phase, the sensor responses in the LT phase were higher across the full range of gas concentrations for both H_2 and NH_3 gases. This could attribute to the entrapment of pre-chemisorbed gas analytes and reactive oxygen species in the potential well^{51,57,58}, as well as the influence of temperature on the electrical conductivity of the sensing channel. In addition, gas responses to H_2 decreased as the duty cycle was reduced due to the slower kinetics of H_2 chemisorption on the MOS surface. In contrast, NH_3 showed an increased response with shorter net reaction times and longer desorption periods, indicating that the surface reaction rate is influenced by varying duty cycles. CO exhibited consistently higher responses in the HT phase than in the LT phase, possibly because of variations in diffusion depths during the different pulse phases. These findings highlight the importance of optimizing the duty cycle (which tailors the adsorption/desorption equilibrium) to enhance the sensor's efficiency and selectivity.

The response of the MOS gas sensor is governed by two key factors: the rate constant (k) and diffusion coefficient (D_k). Thus, Knudsen diffusion is applied to model gas diffusion within the nanosheet sensing layer, and the

mathematical expression for D_k is ref. ⁶⁶:

$$D_k = (4r/3)\sqrt{2RT/\pi M} \quad (1)$$

where r , T , and M are pore radius, temperature, and gas molecular mass, respectively. At a constant temperature, a smaller M results in a larger D_k . Thus, these three gases are ranked with D_k from highest to lowest as H_2 , NH_3 , and CO . The transient conductivity variations can be attributed to differences in the quantity of oxygen re-adsorption, which is dependent on the surface reaction rate. A faster surface reaction rate leads to greater oxygen consumption and reduces the amount of oxygen available for re-adsorption. According to the surface gas sensing kinetic model, the expression for the response time constant t_{res} can be expressed as ^{63,67}:

$$t_{res} = (k_r C_V + k_{-r})^{-1} \quad (2)$$

where k_r represents the surface reaction constant, C_V is the reducing gas concentration, and k_{-r} is the inverse kinetic reaction rate constant. The transient response of the fabricated gas sensor to 50–500 ppm H_2 , CO , and NH_3 gases under LT and HT pulses is used to extract t_{res} . By plotting $1/t_{res}$ as a function of C_V curve, k_r can be estimated (Fig. S4). For instance, the values of k_r for H_2 , CO , and NH_3 under the PHM1-HT mode are 1.10×10^{-5} , 2.76×10^{-6} , and 1.18×10^{-5} , respectively; under the PHM1-LT mode, the values of k_r are 7.01×10^{-6} , 4.50×10^{-6} , and 5.20×10^{-6} . Thus, in the PHM1-LT mode, the gases are ranked in terms of k_r from lowest to highest as CO , NH_3 , and H_2 . When the pulsed operation mode shifts from PHM1 to PHM3 (increasing the duty cycle from 1/3 to 2/3), the k_r value for H_2 decreases, while that for NH_3 increases. Notably, CO consistently exhibits higher k_r values under LT compared to HT, possibly due to its larger M , distinct spatial configuration, and reduced desorption at higher temperatures.

The variations in the k_r and D_k contribute to differences in the diffusion depths of the three gases, which in turn affect the magnitude of the gas responses ^{24,66}. The diffusion depth can be expressed as:

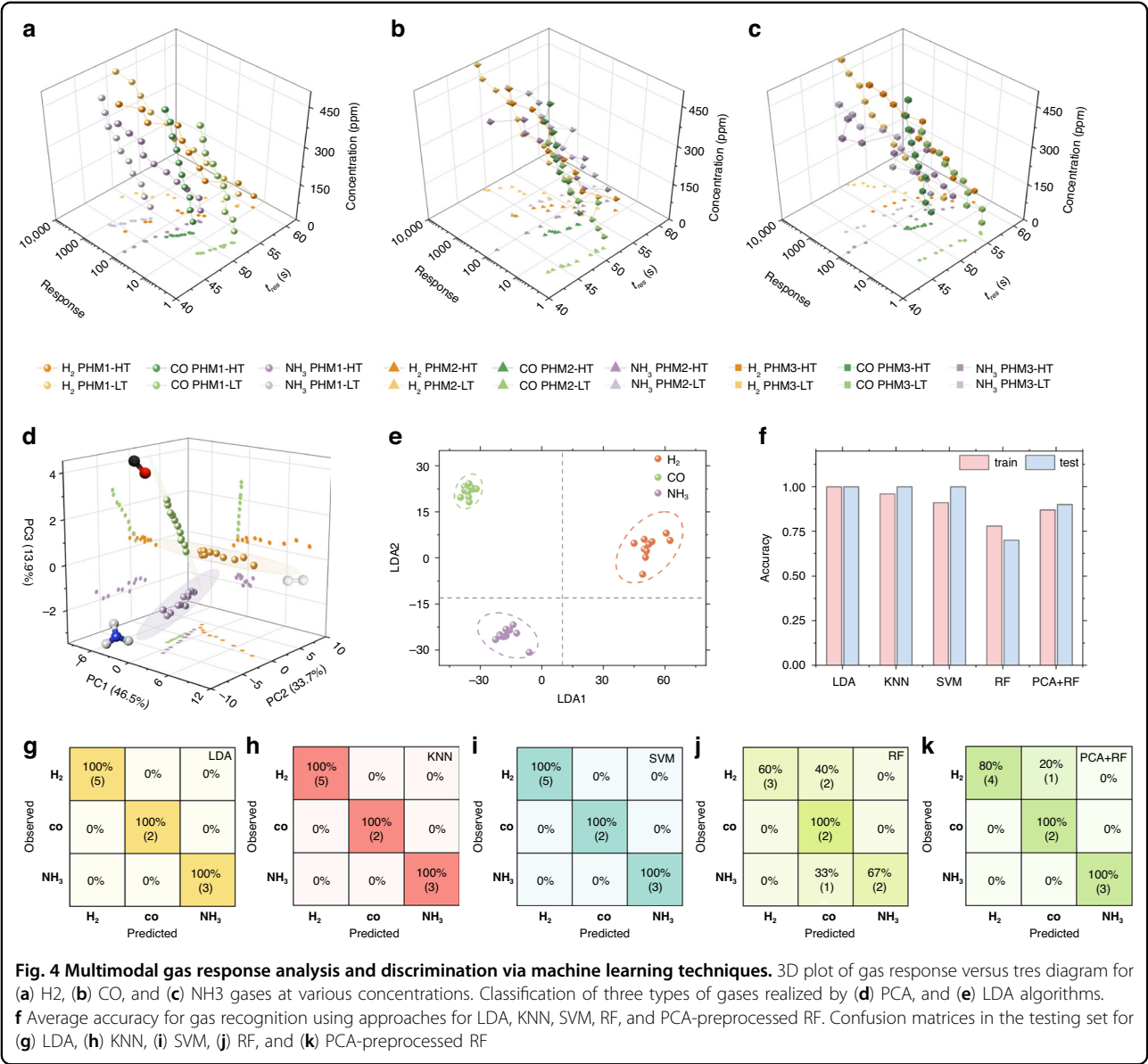
$$C_A/C_{A,s} = \cos h((L-x)\sqrt{k/D_k}) / \cos h(L\sqrt{k/D_k}) \quad (3)$$

where C_A is the gas concentration, $C_{A,s}$ is the concentration at the sensing surface, L is the sensing layer thickness, and x is the distance from the sensing layer surface. When V_H increases from 1 V to 3 V, additional gas molecules may diffuse into the sensing layer, where chemisorption reactions consume adsorbed oxygen species, thereby releasing trapped electrons and increasing conductivity. As shown in Fig. 3d–f, H_2 and NH_3 diffuse deeper into the sensing layer under the PHM3-LT and PHM1-LT modes,

respectively, leading to larger changes in conductivity during a single temperature cycle. In contrast, CO consumes oxygen more rapidly in the LT state, as suggested by its higher k_r values, thus, oxygen from the air is only available for supply near the surface layer, limiting the diffusion depth within the MOS layer.

The multivariable gas-response features attained from a single MEMS gas sensor during its pulsed phase provide valuable insights into surface gas-solid interactions and gas diffusion. These features suggest an opportunity for enhancing gas selectivity and discriminating between different gases through algorithmic analysis. The goal is to predict gas types accurately based on variations in sensing performance, thereby improving sensor robustness in real-world applications where selectivity may be constrained. To evaluate the gas categorization capability of the duty-driven sensor operation, we analyzed 3D plots of gas responses and t_{res} at LT and HT for different gas types and concentrations (Fig. 4a–c). Notably, they appear positively correlated during the pulsed phase. Although a clear trend emerges in response to increasing gas concentration, these plots alone are insufficient for distinguishing gas species and concentrations through simple correlation.

Moreover, we applied a transformation to the data using PCA, projecting it into a new coordinate system where the principal components represent directions of maximum variance ⁶⁸. This transformation minimizes information loss, emphasizing the most relevant features for easier visualization and analysis. In the PCA plots (Fig. 4d), data points are clustered according to gas categories, with principal components PC1 (46.5%), PC2 (33.7%), and PC3 (13.9%). Although the PCA plots provide insight into the extracted gas-sensing features from different PHMs, the data points from the three gases show some overlap, especially at varying concentrations, making it challenging to distinctly identify gas types and concentrations using merely PCA. To enhance the identification process, LDA was utilized, which has proven effective in reducing the dimensionality of response values for gas identification. During the potential application scenario of miniaturized mobile sensing systems, the LDA feature space (Fig. 4e) clearly discriminates the three gases, with distinct color-coded representations. The LDA method is employed to classify data into distinct groups by building a feature space defined by vectors derived from linear combinations of predictor variables. In this study, we applied the LDA technique to process aggregated data sets obtained from a single gas sensor, generating a two-dimensional feature map. One of the data sets represents the percentage variation in 12 distinct gas-sensing features across different concentrations of three single-component gases. When a single-component gas is introduced into the environment, the sensor responds, causing variations in



the gas-sensing features. Each set of these multivariable changes corresponds to a point in the trained LDA feature space, with the gas species identified based on the proximity of the point to the nearest group centroid in this space. As a result, the single gas sensor demonstrated excellent identification capabilities for H₂, CO, and NH₃, achieving a recognition accuracy of up to 100%.

A predictive electronic nose system tends to construct a generalized pattern recognition model during the training phase, based on known data, and then use this model to identify unknown data. To assess the effectiveness of various ML classification techniques for temporal pattern analysis, we employed supervised methods, including LDA, KNN, SVM, and RF. The algorithms were implemented in PyCharm, and the corresponding code is

provided in Tables S2–S6. A k -fold cross-validation approach was employed (a k -fold value of 4). Each classification algorithm was separately trained on the training data and subsequently tested on the unknown dataset. Figure 4f shows the average classification accuracy for all gases across the testing sets. In this work, preprocessing the data with PCA prior to applying the RF algorithm further improved the classification accuracy by reducing dimensionality and highlighting the most significant features. A confusion matrix (Fig. 4g–k) was also used to assess classification accuracy for each gas, providing insights into which gases were most often misclassified. While the LDA, KNN, and SVM classifiers achieved nearly 100% accuracy in predicting H₂, CO, and NH₃ gases within the testing sets, the RF classifier exhibited

lower accuracies in identifying H₂ (60%) and NH₃ (67%), respectively, resulting in frequent misclassifications. This indicates that classification accuracy is primarily influenced by the choice of algorithm and the variations in sensing parameters across different gases. When the gas-sensing dataset is acquired under varying concentrations, exposure times, and complex conditions, it may be necessary to re-assess key aspects of the algorithmic analysis, including feature extraction, algorithm selection, and model development. The objective is to enable assessments of analytes across diverse conditions. The high prediction accuracy using LDA, KNN, and SVM classifiers is consistent with the findings from the LDA feature space analyses, which reveal clear separations among the gas clusters. The well-defined feature space not only supports more accurate classification but also strengthens the robustness of the gas recognition system. The consistent performance across the majority of the algorithms highlights the enhanced discriminative power of the enriched dataset, which includes various PHM modes. This suggests that using a single pulse-driven sensor for identifying different target gases is a viable approach.

Further, Table S7 presents a comparison of the proposed single-sensor-based system with recently developed ML-based gas identification systems. In this study, we introduce a low-power electronic nose system utilizing a pulse-driven, single microheater-embedded MOS sensor. This system demonstrates cost-effective power consumption and classification accuracy comparable to that of state-of-the-art MOS systems. Furthermore, the unique signals generated through pulsed heating, combined with various ML techniques, facilitate gas identification using only a single sensor. This single-sensor approach holds promise for future expansion to identify a broader range of gases, considering distinct diffusion behaviors and sensing kinetics. Additionally, by reducing power consumption, enhancing signal readout, and facilitating molecule recognition through feature extraction, our gas identification strategy offers significant potential for developing highly sensitive, stable, and low-power MOS electronic noses for advanced applications. For future outlooks, we plan to integrate the MOS gas sensor with programmable analog and digital circuits. The interface circuit will facilitate sensor configuration, including features such as a pulsed heating voltage generator, signal acquisition, data processing, and output of recognition results, all driven by embedded timing and algorithmic code. With the integration of a ML classification algorithm within the processor, this single-sensor electronic nose system holds significant potential for accurately identifying various gas types in applications such as indoor air quality monitoring and food spoilage detection.

Conclusions

A dual-mode gas sensor has been developed to resolve the selectivity limitations, demonstrating the ability to identify multiple single-component gases through pulsed temperature operation. The sensor features SnO₂ nanosheets that provide sufficient reaction sites, coupled with an integrated high-efficiency microheater. The fabrication process is compatible with standard MEMS techniques, enabling reductions in device size and facilitating scalable production. By applying repeated pulsed power inputs to the microheater, our approach effectively decouples the influences of temperature, diffusion characteristics, and reaction kinetics on sensing signals. The distinctive features extracted from dual responses are analyzed using multiple ML algorithms, allowing the sensor system to achieve a 100% accuracy rate for three hazardous gases (H₂, CO, and NH₃). The pulse-driven, single-sensor electronic nose system offer a cost-effective and low-power solution, making it promising for mobile sensing and IoT applications.

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Conflict of interest

The authors declare no competing interests.

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